

**COMPUTER STATISTIC
PROBABILITY DISTRIBUTION**



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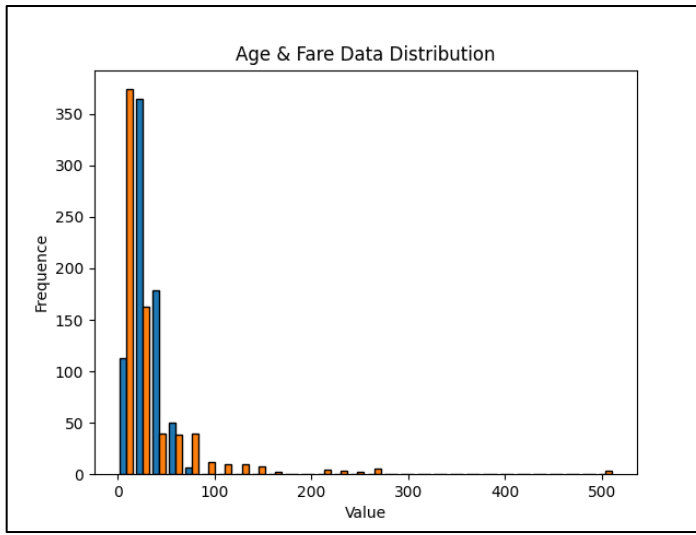
ASSIGNMENT PART 1

- **Age and Fare Missing Value**

```
Age      177
Fare      0
dtype: int64
```

From image above, Age has 177 missing values and Fare not have missing values.

- **Age and Fare Data Distribution**



From the Histogram diagram above, it can be seen that most of the data is concentrated in low values, especially in the range of 0 to 100, with the highest frequency being in the smaller value intervals. The distribution is right-skewed, indicating that there are some values that are much larger than the majority of the data.

- **Simple Data Imputation**

Since the Histogram diagram shows that the data has outliers, I used the median to assign values to the missing values in the Age and Fare variables.

```
import pandas as pd

dataset = pd.read_csv('../Titanic-Dataset.csv')

print("Total missing value before imputation:")
print(dataset[['Age', 'Fare']].isnull().sum())

# Impute missing value
dataset = dataset.assign(
    Age=dataset['Age'].fillna(dataset['Age'].median()),
    Fare=dataset['Fare'].fillna(dataset['Fare'].median())
)

print("Total missing value after imputation:")
print(dataset[['Age', 'Fare']].isnull().sum())
```

```
Total missing value before imputation:
Age      177
Fare      0
dtype: int64
Total missing value after imputation:
Age      0
Fare      0
dtype: int64
```

ASSIGNMENT PART 2

- **Data Imputation with Mean Method**

```
import pandas as pd

dataset = pd.read_csv('../Titanic-Dataset.csv')

print("Total missing value before imputation:")
print(dataset[['Age', 'Fare']].isnull().sum())

# Impute missing value
dataset = dataset.assign(
    Age=dataset['Age'].fillna(dataset['Age'].mean()),
    Fare=dataset['Fare'].fillna(dataset['Fare'].mean())
)

print("Total missing value after imputation:")
print(dataset[['Age', 'Fare']].isnull().sum())
```

7 bin/python 7 users/ khencanyo/ khencanyo/MyCampus
Total missing value before imputation:
Age 177
Fare 0
dtype: int64
Total missing value after imputation:
Age 0
Fare 0
dtype: int64

- **Age Z-Score**

	Age	Age_Zscore
0	22.0	-0.530377
1	38.0	0.571831
2	26.0	-0.254825
3	35.0	0.365167
4	35.0	0.365167

- **Total Outlier from Age with $|Z| > 3$**

```
7 bin/python 7 users/ khencanyo/ khencanyo/MyCampus  
Total outlier in 'Age' with  $|Z| > 3 = 2$   
jobcheat-answers@khencanyo@Khens-MacBook-Air:~$
```

- **Age in Normal Distribution**

```
7 bin/python 7 users/ khencanyo/ khencanyo/MyCampus  
Probability P(Age < 20): 0.2522  
jobcheat-answers@khencanyo@Khens-MacBook-Air:~$
```